MS 6015: Multivariate Statistical Methods for Business

Principal Component Analysis

Outline

Introduction

- Researchers deal with hundreds of variables in their analyses. With that many variables it is difficult to comprehend the patterns of association.
- This is often complicated by the fact that there is often substantial redundancy among dimensions, leading to high levels of correlation and multi-collinearity.
- Principal component analysis is a method for re-expressing multivariate data. It allows the researcher to reorient the data so that the first few dimensions account for as much of the available information as possible.
- Each principal component is uncorrelated with all the others and hence captures independent pieces of the information puzzle represented in the larger data set.
- Providing meaningful interpretation to these principal components is one of the challenges in PCA analysis.

Main Idea

- Reduce the dimensionality of a data set in which there is a large number of inter-related variables while retaining as much as possible the variation in the original set of variables.
- The reduction is achieved by transforming the original variables to a new set of variables, "principal components", that are uncorrelated and ordered such that the first few retains most of the variation present in the data.

Goals

- Create a new set of variables (a smaller number that are uncorrelated). These can be used in other procedures (e.g., multiple regression).
- Select a sub-set of the original variables to be used in other multivariate procedures.
- Detect outliers or clusters of observations.

What Is PCA?

Motivation

- Data with p variables lives in $\mathbb{R}^p$? hard to visualise or model when p is large
- Many variables move together $\Rightarrow$ redundancy
- **PCA** finds a new coordinate system that:
  - Maximises variance along each axis
  - Keeps axes mutually orthogonal
  - Preserves data structure with fewer dimensions

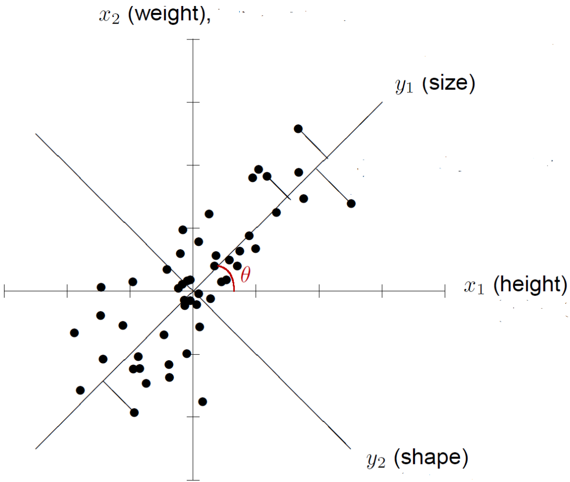
Key Properties

Concept	**Meaning**
Dim. Reduction	p vars $\to$ k PCs, $k \ll p$
Orthogonality	PCs are uncorrelated
Linearity	Each PC is a linear combination
Variance	PCs ordered by variance captured

Result: A small set of uncorrelated *principal components* that explain most of the total variance.

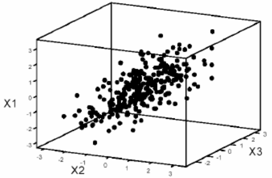
Geometric Intuition

Geometric Intuition



Intuition- Ref Lattin 4.2.1

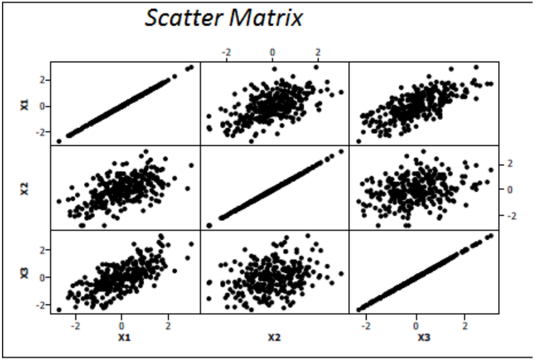
The intuition of PCA is best illustrated with an example based on artificial data.



Correlation Matrix

	X1	X2	X3
X1	1.000		
X2	0.562	1.000	
X3	0.704	0.304	1.000
Var	1.000	1.000	1.000

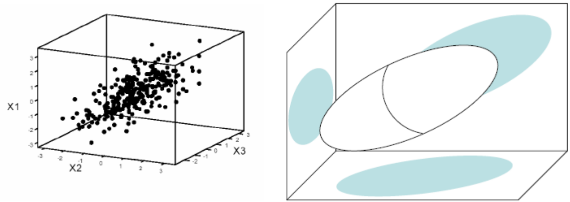
Correlation is largest between X_1 and X_3 and smallest between X_2 and X_3.



Would it be possible to use a single dimension to capture and convey most (if not all) of the information contained in the variables X_1, X_2 and X_3?

This is what principal component analysis is about: Finding a smaller number of variables that accounts for a sufficient amount of the original information.

Stylized view of Scatter plot

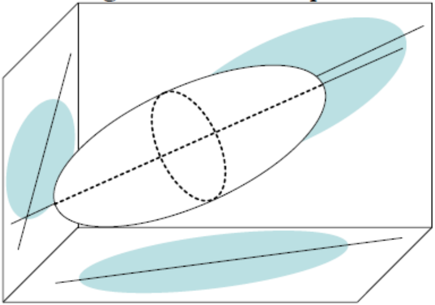


Shadows represent shapes in two dimensions

First Principal Component

Recall that a vector-vector multiplication $v' u$ is a projection of v onto the line spanned by u.

Stylized view of First Principal Component:
Longest Axis of the Ellipsoid.



Let Z_1 be projection of all data points onto the longest axis.

First Principal Component

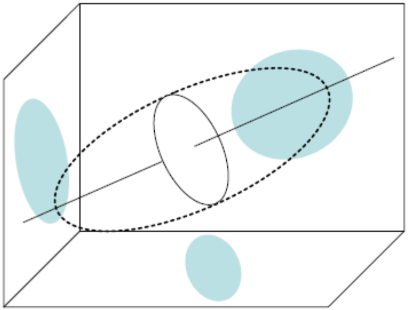
- The variance of this component, its eigenvalue, is 2.063
- In other words it accounts for twice as much variance as any single variable
- Note 3 variables = 2.063/3 = .688% variance accounted for by this first component 1

Total Variance Explained

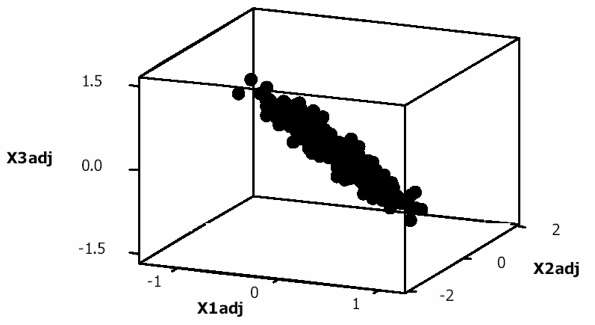
Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.063	68.778	68.778	2.063	68.778	68.778
2	.706	23.518	92.296			
3	.231	7.704	100.000			

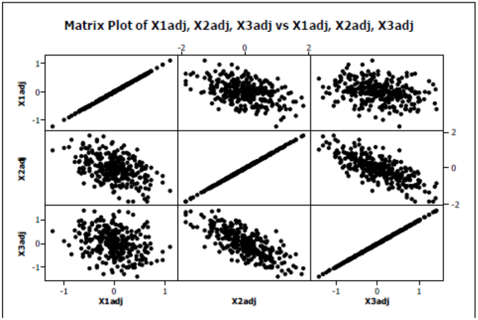
Extraction Method: Principal Component Analysis.

We can now calculate the projection $p(x)$ of each point x onto the line spanned by the vector u(constituting the first principal component). The difference vector $x - p(x)$ is the projection of x onto the plane perpendicular to the vector u_1



Denoting the coordinates of the projected points $x - p(x)$, with X_{1adj}, X_{2adj} and X_{3adj} in a three-dimensional scatter plot we have:

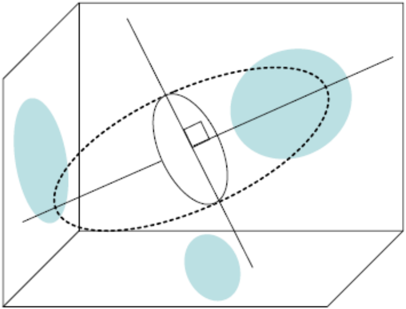




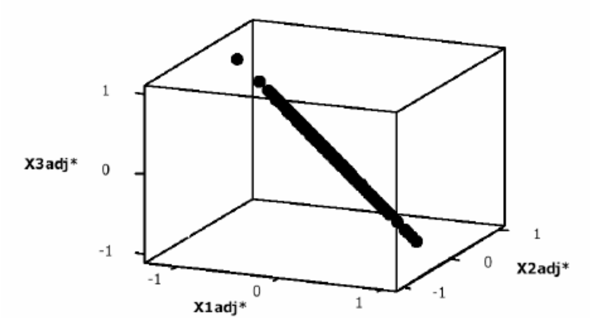
- Observe now that after removing the information of the first principal component Z_1 the remaining association suggests a strong negative correlation between X_2 and X_3.
- This is not obvious from the original correlation matrix and illustrates why PCA helps in understanding more the association about the variables.

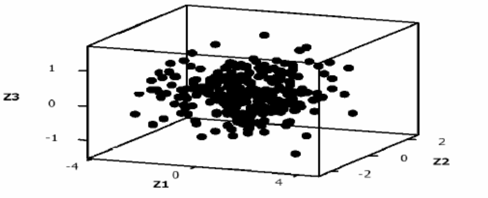
Second Principal Component

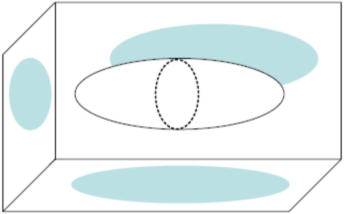
To find the second principal component we are searching for the linear combination of $(X_{1adj}, X_{2adj}, X_{3adj})$ that has now the largest variance.



We can now calculate the projection $p(x)$ of each point x onto the plane spanned by the vectors u_1 and u_2. The difference vector $x - p(x)$ is the projection of x onto the line perpendicular to the vectors u_1 and u_2




1
Z3
0
-1
-4
0
Z1
4
2
0
Z2
-2



Observations

- The new variables Z_1, Z_2 and Z_3 are mutually uncorrelated, and each new variables is chosen to account for the maximum amount of variation not already accounted for by the previously chosen variables.
- Although it takes all three variables to explain for all of the information, in our example Z_1 and Z_2 together account for a variance of 92.3%
- While this dimension reduction is not a big deal here, it is easy to imagine the benifits of being able to reduce 20 variables down to four or five dimensions.

Component Loadings

- It is usually instructive to examine the relationship between Z_1, Z_2 and Z_3 and the original variables X_1, X_2 and X_3 One way to do this is look at their correlations.
- The correlations between $(X_1, Z_1), (X_2, Z_1), (X_3, Z_1)$ are called the principal component loadings of Z_1 onto the original variables. If all these correlations are high, then Z_1 reflects a component of shared variance underlying all three original variables.

Component Loadings

- The component loadings can inform us as to their interpretation
- They are the original variable's correlation with the component
- In this case, all load nicely on the first component, which since the others do not account for nearly as much variance is probably the only one to interpret
- Depending on the type of PCA, the rotation etc. you may see different loadings although often the general pattern will remain

Component Loadings

Component Matrix[a]

	Component		
	1	2	3
col1	.928	-.080	-.364
col2	.726	.670	.159
col3	.822	-.501	.271

Extraction Method: Principal Component Analysis.

a. 3 components extracted.

Two Approaches

- Algebraically:PCs are linear combinations of p original variables $X_1, X_2, \ldots, X_p$ such that
  - The first PC has the largest variance as possible,
  - The second PC has the largest variance as possible and is orthogonal to the first
- Geometrically:
  - Rotation to a new coordinate system.
  - "Best" fit hyper-plane.

Suppose that we have a random vector $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$ with variance covariance matrix

$$\mathrm{var}(\mathbf{X}) = \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \dots & \sigma_p^2 \end{pmatrix}$$

- We want to transform p variables; $X_1, X_2, \ldots, X_p$ to q orthogonal linear combinations where $q << p$
- Let $X^{'} = (X_1, X_2, \ldots, X_p)$ have the covariance matrix Σ with eigen values $\lambda_1 \geq \lambda_2 \geq \ldots, \lambda_p \geq 0$
- Consider p linear combinations

$$Y_i = a_i^{'} X = a_{i1} X_1 + a_{i2} X_2 + \ldots + a_{ip} X_p$$

$i = 1, 2, \ldots, p$

  - $Var(Y_i) = a_i^{'} \Sigma a_i; i = 1, 2, \ldots, p$
  - $Cov(Y_i, Y_k) = a_i^{'} \Sigma a_k; i, k = 1, 2, \ldots, p$

Principal Components

PCs are the uncorrelated linear combinations, $cov(Y_i, Y_k) = 0$ for all $i \neq k$, with variances as large as possible.

- First principal component is the linear combination with maximum variance. That is $max \;\; Var(Y_1) = max \;\; a_1^{'}\Sigma a_1$
- To eliminate indeterminacy, restrict attention to unit vectors. Therefore
  - First principal component is the linear combination $a_1^{'}X$ which maximises $max \;\; Var(Y_1) = max \;\; a_1^{'}\Sigma a_1$ subject to $a_1^{'}a_1 = 1$.
  - Second principal component is the linear combination $a_2^{'}X$ which maximises $max \;\; Var(Y_2) = max \;\; a_2^{'}\Sigma a_2$ subject to $a_2^{'}a_2 = 1$ and $Cov(a_1^{'}X, a_2^{'}X) = 0$.
  - i^{th} principal component is the linear combination $a_i^{'}X$ which maximises $max \;\; Var(Y_i) = max \;\; a_i^{'}\Sigma a_i$ subject to $a_i^{'}a_i = 1$ and $Cov(a_i^{'}X, a_k^{'}X) = 0$ for $k < i$.

Maximizing Criteria- Result 8.1(JW)

Theorem

Let Σ be the covariance matrix associated with $X^{'} = (X_1, X_2, \ldots, X_p)$. Let Σ have the eigen value- eigen vector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \ldots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \ldots \lambda_p \geq 0$. Then the i^{th} principal component is given by $Y_i = \mathbf{e}_i^{'} X$ with $Var(Y_i) = \mathbf{e}_i^{'} \Sigma \mathbf{e}_i; i = 1, 2, \ldots, p$ and $Cov(Y_i Y_k) = \mathbf{e}_i^{'} \Sigma \mathbf{e}_k = 0; i \neq k$

Proof.

From linear algebra, we know that

$$max_{a \neq 0} \left(\frac{a^{'} \Sigma a}{a^{'} a} \right) = \lambda_1 = \frac{e_1^{'} \Sigma e_1}{e_1^{'} e_1} = e_1^{'} \Sigma e_1 = Var(Y_1)$$

□

Total Population Variance

Let $X^{'} = (X_1, X_2, \ldots, X_p)$ have covariance matrix Σ with eigen value-eigen vector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \ldots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \ldots \lambda_p \geq 0$. Let $Y_i = e_i^{'} X; i = 1, 2, \ldots, p$ be the principal components. Then

$$\sigma_{11} + \sigma_{22} + \ldots + \sigma_{pp} = \lambda_1 + \lambda_2 + \ldots + \lambda_p$$

The Total Population Variance is preserved by the transformation. The Proportion of total variance due to the k^{th} PC is

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \ldots + \lambda_p} = \frac{k}{\sum_{i=1}^{p} \lambda_i}; k = 1, 2, , \ldots, p$$

Proportion of Variance accounted for

- We often select q PCs such that the proportions for $k = 1, ..., q$ sum up as close to 1 (yet not too large of a value for q).
- The Proportion of Variance accounted for by the first q PCs $= \frac{\sum_{k=1}^{q} \lambda_k}{trace(\Sigma)}$
- We try to balance the percent of variance (information) retained and the number of PCs (simplicity).
- Often we are interested interpreting the new variables (i.e., the PCs), so we examine the elements of the $\mathbf{e}_i^{'}s$
- The size (magnitude) of the elements of $\mathbf{e}_i$ are an indicator of a variable's "importance" to the i^{th} PC.

Correlation between Y_i and X_k

If $Y_i = \mathbf{e}_i^{'} X; i = 1, 2, \ldots, p$ are the principal components obtained from Σ, then

$$\rho_{Y_i, X_k} = \frac{e_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}; i, k = 1, 2, \ldots, p$$

Example

- Random Variables, X_1, X_2, X_3 have covariance matrix

$$\Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

- Eigen value-eigen vector pairs
$\lambda_1 = 5.83, e_1^{'} = (0.383, -0.924, 0)$
$\lambda_2 = 2.00, e_2^{'} = (0, 0, 1)$
$\lambda_3 = 0.17, e_3^{'} = (0.924, 0.383, 0)$
- Principal Components:
$Y_1 = 0.383X_1 - 0.924X_2$
$Y_2 = X_3$
$Y_3 = 0.924X_1 + 0.383X_2$

Questions

- $$Var(Y_1) = ?? =$$
- $$Cov(Y_1, Y_2) = ??$$
- Total population variance??
- Proportion of total variance accounted for by first principal component= ?
- Calculate and Interpret $\rho_{Y_1 X_1}, \rho_{Y_1, X_2}, \rho_{Y_2, X_1}, \rho_{Y_2, X_2}$ and ρ_{Y_2, X_3}

Step 1: Data & Centering

Step 1 - Data Matrix

Collect n observations on p variables:

$$X \in \mathbb{R}^{n \times p}$$

Rows = observations, Columns = variables (features)

Step 2 - Centre the Data

Subtract the column mean $\mu_j = \frac{1}{n} \sum_i X_{ij}$ from every entry:

$$\tilde{X} = X - \mathbf{1}\,\boldsymbol{\mu}^\top$$

Centring ensures the origin coincides with the data cloud's centre of mass.

Step 3 - Covariance Matrix

Form the $p \times p$ symmetric positive semi-definite sample covariance matrix:

$$\Sigma = \frac{1}{n-1}\,\tilde{X}^\top \tilde{X}$$

Step 2: Eigendecomposition

Step 4 - Solve the Eigenvalue Problem

Find all p eigenpairs $(\lambda_j, \mathbf{v}_j)$ of Σ:

$$\Sigma\,\mathbf{v}_j = \lambda_j\,\mathbf{v}_j, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$$

Stacking eigenvectors as columns gives the **spectral decomposition**:

$$\Sigma = V\,\Lambda\,V^\top$$

where $V = [\mathbf{v}_1 \cdots \mathbf{v}_p]$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$.

Step 5- PC Scores

Project data onto the first k eigenvectors:

$$Z = \tilde{X}\,V_k$$

$V_k \in \mathbb{R}^{p\times k}$: first k columns of V
$Z \in \mathbb{R}^{n\times k}$: **score matrix**

Variance Explained by PC j

$$\text{Prop}_j = \frac{\lambda_j}{\sum_{i=1}^{p} \lambda_i} \qquad \text{Cumulative}_k = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{i=1}^{p} \lambda_i}$$

Factor Loadings-Definition & Properties

Definition

The factor loading v_{ij} is the (i, j) entry of V - the weight of original variable i in principal component j. Each loading equals the *correlation* between variable X_i and PC Z_j (when data are standardised).

Properties

Property	**Detail**
Range	-1 to $+1$
Orthogonality	$V^\top V = I$
Sign	Only relative sign matters
Magnitude	$\lvert v_{ij} \rvert$ large $\Rightarrow X_i$ drives PC$_j$

Interpretation

- **Large $+$ve** - variable moves *with* the PC
- **Large $-$ve** - variable moves *opposite* to PC
- Near zero - variable barely contributes
- Naming PCs: group high-loading variables and give economic meaning

Factor Loadings - Example (3 Variables, 3 PCs)

Loading Matrix

Var	Z1	Z2	Z3
X1	**0.928**	−0.080	−0.364
X2	0.726	**0.670**	0.159
X3	0.822	−0.501	0.271
Var	2.063	0.706	0.231
% Expl.	68.8%	23.5%	7.7%
Cum %	68.8%	92.3%	100.0%

% Variance by Variable

Var	Z1	Z2	Z3
X1	**86.1**	0.6	13.2
X2	52.6	44.8	2.5
X3	67.6	25.1	7.3
X1 cum	86.1	86.8	**100**
X2 cum	52.6	97.5	**100**
X3 cum	67.6	92.7	**100**

Key calculation

$86.1\% = (0.928)^2$

$7.7\% = 0.231/3$

Additive because...

Z1, Z2, Z3 are **uncorrelated**

Row sums 100%

Var. of each standardised $X_i = 1$

Factor Plot (Biplot)

What Is a Biplot?

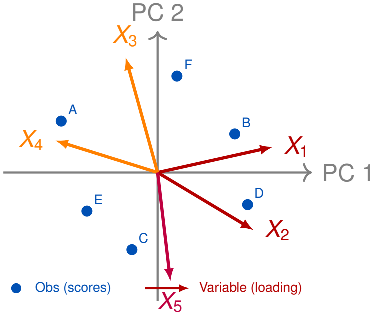
A single plot that overlays:

- **Scores** - observation points in PC space
- **Loading vectors** - arrows whose direction and length show correlation with the PCs

Reading a Biplot

- Arrow **angle** $\approx$ correlation between variables
- Arrow **length** $\propto$ variance explained

Schematic Biplot



Example: Bank data

- Suppose a bank collects the following eight variables for loan applicants: Income, Education, Age, Residency at current address, Years at current employer, Savings, Debt, and the number of credit cards.
- Use principal component analysis to see if we can reduce the number of dimensions yet retain most of the information.

GSP: Case study

1. Describe the dataset.
2. GSP raw data: What do the correlations indicate? Which single component accounts for most variation.
3. GSP_share: How different is this from GSP_raw?
4. What information do loading plots give us? Comment on the outliers.

Gross State Product-Data

- Data (expressed in millions of dollars) are measures of Gross State Products (GSP) for each of 13 different areas of economic activity in 1996 for 50 states in the U.S.

Index	Description	Index	Description
1	Agriculture	8	Utilities
2	Mining	9	Whole-Sale
3	Construction	10	Retail
4	Manufacturing-Durable Goods	11	Fiduciary
5	Manufacturing-Nondurable Goods	12	Service
6	Transportation	13	Government
7	Communication		

Correlaton Matrix

Correlation Matrix of GSP among 13 measures of economic activity (raw data)

	Agricultu re	Mining	Constr-uction	Manu-Dur	Man-Nondur	Trans-portation	Commun ic.	Utilities	Whole-Sale	Retail	Fidu-ciary	Service	Govt
Agriculture	1.000												
Mining	0.248	1.000											
Construction	0.804	0.415	1.000										
Manu-Dur	0.749	0.262	0.873	1.000									
Man-Nondur	0.662	0.391	0.879	0.841	1.000								
Transportation	0.813	0.458	0.976	0.862	0.896	1.000							
Communic.	0.716	0.356	0.930	0.742	0.835	0.923	1.000						
Utilities	0.674	0.525	0.951	0.837	0.907	0.942	0.903	1.000					
Whole-Sale	0.815	0.346	0.977	0.877	0.883	0.973	0.953	0.930	1.000				
Retail	0.848	0.343	0.984	0.886	0.862	0.965	0.929	0.920	0.984	1.000			
Fiduciary	0.740	0.190	0.902	0.790	0.793	0.876	0.946	0.851	0.944	0.933	1.000		
Service	0.804	0.269	0.955	0.846	0.829	0.932	0.950	0.894	0.979	0.978	0.983	1.000	
Govt	0.814	0.344	0.972	0.843	0.868	0.949	0.957	0.915	0.974	0.982	0.949	0.978	1.000

Correlations of higher than 0.80 are indicated in color.

Observations from Data and Correlation Matrix

- One would expect a lot of positive correlation because one would expect high levels of GSP across all industries for the largest and the most prosperous states.
- For this data it may be possible that a good deal of the covariation can be explained by a single principal component, being, the size of each state- Check!!

Eigen Values

Total Variance Explained

Component	Initial Eigenvalues		
	Total	% of Variance	Cumulative %
1	10.944	84.187	84.187
2	.979	7.534	91.721
3	.400	3.077	94.799
4	.343	2.636	97.435
5	.139	1.071	98.505
6	.069	.534	99.039
7	.040	.309	99.348
8	.033	.256	99.604
9	.025	.193	99.798
10	.011	.081	99.879
11	.007	.058	99.936
12	.006	.047	99.983
13	.002	.017	100.000

Extraction Method: Principal Component Analysis.

Proportion of Variance Explained

Total Variance = 13
Variance accounted for by

Z_1	Z_2	Z_3
10.9443	0.9794	0.4001

- Note that component Z_1 explains 84.2% of variance in the original data.
- The correlation of Z_1 with the population size of a state happens to be 0.995
- The revenue of mining apparently is not linked to population size.

Other Questions

- It may also be of interest to look at the share (or percentage) that each industry contributes to the total GSP of a particular state.
- For example do some states generate a greater proportion of their GSP from mining than from agriculture?
- Are certain areas of economic activity (eg. retail and wholesale) positively correlated?

Correlation Matrix for Shares

Correlation Matrix of GSP of 13 measures of economic activity (normalized data)

	Agriculture	Mining	Constr-uction	Manu-Dur	Man-Nondur	Trans-portation	Communic.	Utilities	Whole-Sale	Retail	Fidu-ciary	Service	Govt
Agriculture	1.000												
Mining	-0.061	1.000											
Construction	0.083	-0.025	1.000										
Manu-Dur	0.030	-0.423	-0.123	1.000									
Man-Nondur	-0.145	-0.138	-0.317	0.203	1.000								
Transportation	0.278	0.615	0.073	-0.356	-0.176	1.000							
Communic.	-0.182	-0.195	-0.034	-0.325	-0.101	-0.052	1.000						
Utilities	0.044	0.396	0.021	-0.059	0.070	-0.044	-0.175	1.000					
Whole-Sale	0.245	-0.548	-0.088	0.267	0.039	-0.211	0.330	-0.273	1.000				
Retail	0.092	-0.392	0.407	0.191	-0.126	-0.145	0.120	0.041	0.162	1.000			
Fiduciary	-0.300	-0.407	-0.255	-0.183	-0.131	-0.504	0.129	-0.387	0.037	-0.309	1.000		
Service	-0.326	-0.462	0.321	-0.158	-0.456	-0.425	0.312	-0.310	0.239	0.207	0.518	1.000	
Govt	0.113	0.229	0.185	-0.414	-0.239	0.425	0.195	0.052	-0.344	0.288	-0.350	-0.178	1.000

Cells with an $|correlation|$ greater than 0.3 are indicated in color.

Observations

- Note that the entries are not as highly correlated. Whereas the correlation of the raw data are all positive some are negative.
- Most in range from 0 to 0.3; although some are as high as 0.5 to 0.6.
- This suggests that a single component might not be sufficient to account for all underlying variation.
- If it is the case that different states "specialize" mainly in different combinations of economic activity, then we can rely on principal components analysis to help identify these patterns.

Observations Continued

- It is important to note that the correlation matrix of the normalized data is not of full rank, since the data columns now need to sum up to a vector of ones column-wise.
- Thus even though there are 13 variables, the rank of the correlation matrix will be 12.
- The last principal component for the standardized data will have an eigenvalue exactly equal to zero.

Eigen Values

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	3.236	24.889	24.889	3.236	24.889	24.889
2	2.236	17.204	42.092	2.236	17.204	42.092
3	1.960	15.076	57.168	1.960	15.076	57.168
4	1.360	10.464	67.632	1.360	10.464	67.632
5	1.157	8.903	76.535	1.157	8.903	76.535
6	.868	6.679	83.215			
7	.724	5.573	88.788			
8	.616	4.737	93.524			
9	.318	2.448	95.972			
10	.235	1.810	97.783			
11	.152	1.167	98.949			
12	.136	1.050	99.999			
13	6.86E-005	.001	100.000			

Extraction Method: Principal Component Analysis.

Loadings

Component Matrix[a]

	Component				
	1	2	3	4	5
col2	.243	-.011	.539	.435	-.442
col3	.845	-.002	-.364	-.076	-.069
col4	.063	.588	.360	-.408	-.211
col5	-.330	-.562	.526	-.171	-.120
col6	-.017	-.686	.050	.053	.501
col7	.753	.220	.009	.426	-.153
col8	-.273	.472	-.115	.400	.592
col9	.444	-.206	.096	-.485	.217
col10	-.567	-.042	.406	.515	.011
col11	-.162	.390	.710	-.265	.271
col12	-.653	.045	-.625	.014	-.186
col13	-.683	.574	-.178	-.213	-.135
col14	.520	.551	.120	.090	.317

Extraction Method: Principal Component Analysis.

a. 5 components extracted.

Proportion of Variance Explained

Total Variance = 13
Variance accounted for by

Z_1	Z_2	Z_3	Z_4	Z_5
3.236	2.236	1.96	1.36	1.15

- Note that component $Z_1 - Z_3$ explains 57.2% of variance in the original data.
- Component Z_1 is negatively correlated with Mining, Transportation and Govt and positively correlated with Whole-Sale, Service and Fiduciary. Component
- Z_2 is negatively correlated with Manufacturing (both durable and non-durable)and positively correlated with Construction, Service and Government.

Correlation between Z_i and X_k

If $Z_i = \mathbf{e}_i^{'} X; i = 1, 2, \ldots, p$ are the principal components obtained from Σ, then

$$\rho_{Z_i, X_k} = \frac{e_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}; i, k = 1, 2, \ldots, p$$

Scree Plot

The approach involves plotting the variance accounted for each principal component in order from largest to smallest.
Look for an elbow in the curve.

Kaiser's Rule

When scree plot is not a diagnostic, Kaiser's rule is handy
Kaiser recommended retaining only those principal components with eigen values exceeding 1.

Explained Variance

Sometimes it will be important to retain a sufficient number of principal component to be able to account for a certain amount of variance in each of the original variables

When to use PCA

- Until this point we have been assuming that the data is correlated enough to justify dimension reduction using principal component analysis. But what is high enough?
- Bartlett's Sphericity Test: Addresses the question whether the correlation matrix should be factored at all

$$\chi^2\left[\frac{(p^2-p)}{2}\right] = -\left[(n-1)-\frac{(2p+5)}{6}\right] ln|R|$$

where

- $ln|R|$= Natural logarithm of determinant of R
- $\frac{(p^2-p)}{2}$ = Number of degrees of freedom associated with the chi-square test statistic.
- p = the number of variables
- n= the number of observations.

Intuition

The determinant of the correlation matrix is a generalized measure of variance. It can be calculated by taking product of eigen values

$$|R| = \Pi_{j=1}^{p} \lambda_j$$

- When variables truly mutually independent; R approaches I; $|R|$ is close to 1 and $ln|R|$ is close to 0.
- As level of correlation increases, product of eigen values get closer to zero, and $ln|R|$ becomes a large negative number.

- **GSP raw data:** $p = 13,\ n = 50,$

$$|R| = \prod_{j=1}^{13} \lambda_j \approx 5.0e-16,\ -Ln|R| \approx 35.2,\ (n-1) - \frac{(2p+5)}{6} \approx 43.83$$

$$\chi_0 = -\left[(n-1) - \frac{(2p+5)}{6}\right] Ln|R| \approx 1544.07,\ \nu = (p^2 - p)/2 = 78$$
$$p - value = Pr(\chi_v^2 > \chi_0) \approx 5.5e-217 \Rightarrow Reject\ H_0$$

- **GSP normalized data :** $p = 12,\ n = 50$. We only use the first 12 eigenvalues and lower the number of variables by 1 (since they need to sum up to 1).

$$|R| = \prod_{j=1}^{12} \lambda_j \approx 0.013,\ -Ln|R| \approx 4.38,\ (n-1) - \frac{(2p+5)}{6} \approx 44.16$$

$$\chi_0 = -\left[(n-1) - \frac{(2p+5)}{6}\right] Ln|R| \approx 191.6,\ \nu = (p^2 - p)/2 = 66$$
$$p - value = Pr(\chi_v^2 > \chi_0) \approx 3.57e-14 \Rightarrow Reject\ H_0$$

Limitations of Bartlett's

- Problem with Bartlett's test is that for any data set of reasonable size, uncorrelatedness is rejected which here may imply that PCA is appropriate.
- Hence, the decision on whether to use PCA should also be based on the need for dimension reduction in the data set and not only based on a statistical test.
- Principal component analysis is usually conducted by using the sample correlation matrix which can be obtained by calculation the sample covariance matrix of standardized data.

- Principal component analysis may also be conducted on the sample covariance matrix without standardization of the data. However, in this case the underlying variances of the variables may greatly affect that linear combination of the original data columns that achieves maximum variance.
- When using the sample covariance matrix for PCA using non-standardized data one should be able to motivate why the scale information of the original variables need to be retained.

Kaiser-Meyer-Olkin (KMO) Test

- Kaiser-Meyer-Olkin (KMO) Test is a measure of how suited your data is for Factor Analysis.
- The test measures sampling adequacy for each variable in the model and for the complete model.
- The statistic is a measure of the proportion of variance among variables that might be common variance.
- The lower the proportion, the more suited your data is to Factor Analysis.
  - Rule of thumb: If KMO > 0.5 sample is adequate and we can proceed with PCA.

Sample principal components from standardized data

- The weekly m rates of return for five stocks (JP Morgan, Citibank,Wells Fargo, Royal Dutch Shell, and ExxonMobil) listed on the New York Stock Exchange were determined for the period January 2004 through December 2005.
- The weekly rates of return are defined as (current week closing price?previous week closing price)/(previous week closing price), adjusted for stock splits and dividends.
- The observations in 103 successive weeks appear to be independently distributed, but the rates of return across stocks are correlated, because as one expects, stocks tend to move together in response to general economic conditions.

Interpretation

- The first component is a roughly equally weighted sum, or “index," of the five stocks. This component might be called a general stock-market component, or, simply, a market component.
- The second component represents a contrast between the banking stocks (JP Morgan, Citibank,Wells Fargo) and the oil stocks (Royal Dutch Shell, Exxon- Mobil). It might be called an industry component.
- Thus, we see that most of the variation in these stock returns is due to market activity and uncorrelated industry activity.